

Sentiment Fusion: A Sequential Probabilistic Graphical Model for Market Regime Detection and Causal Inference

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Abstract

Traditional quantitative models in finance often process market reactions without identifying the underlying qualitative catalysts. This research proposes the Sentiment Fusion architecture, a hybrid Probabilistic Graphical Model (PGM) that integrates unstructured textual data into a structured market regime framework. Over an 11-year dataset (2012 - 2022) of general financial news, I utilized Latent Dirichlet Allocation (LDA) for thematic feature extraction and a Gaussian Hidden Markov Model (HMM) for sequential state detection. Through D-separation refinement and the analysis of Beta influence coefficients, I successfully stratified the S&P 500 into three distinct regimes, identifying a high-volatility Crisis state at 23.1% annualized volatility. This report details the mathematical architecture, the experimental results, and the personal evolution of my understanding regarding causal inference versus purely statistical machine learning.

1 Introduction

Quantitative forecasting models, such as Autoregressive Integrated Moving Average (ARIMA) and deep learning variants like Long Short-Term Memory (LSTM) networks, have dominated financial time-series analysis for the last decade. These models predominantly rely on price-action history. However, these architectures suffer from a critical flaw: they are inherently reactive black boxes. They fail during Black Swan events or narrative driven volatility because they map historical correlations but fundamentally lack a mechanism for causal inference.

To illustrate this, consider the global market crash of March 2020. An LSTM observing the S&P 500 dropping simply sees a downward numerical vector and projects that momentum forward. It does not know that a global pandemic is the root cause. When a central bank announces a massive stimulus package, the LSTM cannot adjust its trajectory until the price *already* begins to rise. It cannot contextualize the news that causes market panic or euphoria.

1.1 Related Work and Historical Context

The quest to integrate qualitative context into quantitative finance builds upon several foundational pillars of prior research.

Hamilton (1989) revolutionized time-series econometrics by introducing Markov-switching models to detect non-stationary business cycles and structural market regimes . While groundbreaking, Hamilton’s original framework relied exclusively on numerical economic indicators, lacking the ability to parse the unstructured textual narratives that often trigger those regime shifts.

To address this qualitative gap, Tetlock (2007) pioneered the quantitative analysis of financial news. His research demonstrated that media pessimism profoundly impacts trading volume and downward market prices, proving that news is not just a reflection of the market, but a driver of it . Bollen, Mao, and Zeng (2011) extended this paradigm into the digital age by proving that public mood extracted from unstructured social media text (Twitter) could actively predict Dow Jones directional movements with high accuracy . These seminal works established that text is a leading indicator of price action. However, they largely utilized simple lexicon-based sentiment scores (e.g., counting good vs. bad words) rather than modeling the complex causal dependencies between specific, distinct narratives.

More recently, the maturation of Probabilistic Graphical Models (PGMs) has provided the mathematical framework necessary to move beyond mere correlation . Concurrently, Lopez-Lira and Tang (2023) highlighted the modern capacity of Large Language Models (LLMs) to forecast stock price movements by acting as highly advanced, context-aware semantic processors.

1.2 The Sentiment Fusion Architecture

This project establishes a hybrid architecture—Sentiment Fusion—designed to bridge the gap between unstructured qualitative narratives explored by Tetlock, and the structured graphical inference championed by modern PGM theory.

By treating news topics as parent nodes in a Directed Acyclic Graph (DAG), I aimed to identify the causal drivers of volatility across a full decade of financial history. Rather than asking a model to simply predict what the market will do next based on momentum, this framework asks the model to understand why the market is shifting based on the prevailing global narrative. By the end of this pipeline, the PGM acts as a Glass Box, allowing humans to audit the exact mathematical weight of specific news narratives on global market volatility.

2 Data Acquisition

I expanded the scope of the study to encompass the period from 2012 to 2022. This 11-year duration was explicitly selected to capture diverse economic environments, including the Eurozone debt crisis, the 2016 and 2020 US election cycles, and the massive structural shock of the global COVID-19 pandemic.

The primary dataset utilized was sourced from a FinancialNews repository, structured as annual JSON arrays. The raw data contained foundational metadata, including the `headline`, `short_description`, and the source `link` for each daily financial news event.

However, probabilistic models require dense evidence to make accurate inferences. Headlines alone are too sparse to build robust word distributions. To ensure the probabilistic model had sufficient textual depth, I built a dynamic web-scraping pipeline using the `trafilatura` library.

This pipeline visited the provided URLs and extracted the full article bodies. I then fused the headlines, short descriptions, and full scraped text into a single, comprehensive text corpus for each trading day.

3 Conceptual Background & Unsupervised Feature Extraction

Before feeding data into a causal graph, subjective text must be transformed into objective mathematics. Text is inherently unstructured, making it incompatible with standard Bayesian networks that require discrete or continuous numerical states.

3.1 Latent Dirichlet Allocation (LDA)

To solve this, I employed Latent Dirichlet Allocation (LDA). Conceptually, LDA is a generative probabilistic model that assumes every document is created through a mixture of hidden (latent) themes, and that every theme is a mixture of words. By reversing this generative assumption through inference, we can look at a daily corpus of news and mathematically extract the exact percentage weight of various topics discussed that day.

3.2 The Beta (β) Dirichlet Prior

In the generative process of LDA, the parameter β acts as a Dirichlet prior that controls the sparsity of the topic-word distribution. Mathematically, it governs how likely a word is to belong to a specific topic.

A standard uniform distribution would assume words are evenly spread everywhere, which creates blurry topics. By tuning to a lower β value, I forced the model to create highly distinct, specialized topics (e.g., isolating healthcare words from political words). The joint probability distribution of the hidden and observed variables is defined as:

$$P(W, Z, \theta, \phi | \alpha, \beta) = \prod_{k=1}^K P(\phi_k | \beta) \prod_{d=1}^M P(\theta_d | \alpha) \prod_{n=1}^{N_d} P(Z_{d,n} | \theta_d) P(W_{d,n} | \phi_{Z_{d,n}}) \quad (1)$$

Where $\phi_k \sim \text{Dirichlet}(\beta)$ represents the word distribution for topic k . I utilized Variational Bayes inference to optimize this distribution and rapidly extract the topic weights for each trading day.

3.3 LLM-Driven Semantic Topic Mapping

A limitation of LDA is that it outputs mathematical clusters of words without human context (e.g., Topic 0: pay, country, work). To interpret these mathematical outputs, I utilized an LLM as an automated semantic expert. By analyzing the top clustered words for each topic, the LLM generated professional semantic titles and identified specific market signals. Table 1 details these 10 distinct narratives, which serve as the Parent Nodes in the subsequent graphical models.

4 Approach A: Bayesian Belief Network

With the features extracted, I built a Bayesian Belief Network. In Approach A, I shifted the conceptual use of the β parameter from a generative prior (in the LDA phase) to a **Discriminative**

Table 1: LLM-Generated Topic Semantics and Market Signals

ID	Top Words	Professional Name	Market Signal
0	pay, country, work	Macro Labor & Policy	Signal for inflation and employment data.
1	trump, president, election	Executive Political Risk	Primary driver of trade and regulatory volatility.
2	style, wedding, dress	Consumer Discretionary	Signal for retail sector and fashion trends.
3	food, eat, holiday	Hospitality & Leisure	Signal for travel stocks and food staples.
4	video, film, star, movie	Entertainment & Media	Signal for communication services.
5	child, school, parent	Socio-Demographics	Signal for housing and long-term consumer trends.
6	woman, life, coronavirus	Crisis & Global Health	The "Black Swan" node (2020-2022 volatility).
7	city, Hillary, NY	Legacy Urban Politics	Signal for the 2016 cycle and infrastructure.
8	life, way, think, good	General Sentiment (Noise)	Broad "mood" of the general population.
9	police, law, biden, case	Justice & Regulatory	Signal for legal risk and the 2020-present cycle.

Causal Influence Coefficient.

Here, β_i represents the weight or strength of a specific topic's influence on the probability of a future market trend. I modeled the log-odds of a market move Y (Down, Neutral, or Up) taking place the day after a news event:

$$\text{logit}(P(Y = k)) = \beta_0 + \sum_{i=1}^{10} \beta_i \cdot \text{Topic}_i \quad (2)$$

In this formula, a positive β value for the Up class indicates that the presence of that topic causally increases the model's belief that the market will rise. To smooth out daily noise and improve the signal-to-noise ratio, I applied a 5-day rolling average to the text evidence before feeding it into the Bayesian classifier.

5 Approach B: Sequential Gaussian HMM

The limitation of Approach A is that it treats every day as an isolated event. However, financial markets exhibit the Markov Property: the state of the market today is highly dependent on the state of the market yesterday.

To address this, I implemented a Gaussian Hidden Markov Model (HMM) where the true market Hidden State (S_t) is an unobserved regime (e.g., Bull, Bear, or Neutral) that emits observable variables (price volatility and news text).

To train this model unsupervised, I utilized the **Baum-Welch algorithm** (an application of Expectation-Maximization) to iteratively learn the transition probabilities between states and

the Gaussian emission parameters for each state. Once the model learned what a Crisis state looked like probabilistically, I applied the **Viterbi algorithm**. This dynamic programming algorithm allowed me to decode the historical data, finding the single most likely path of hidden regimes the market traversed over the 11-year period given the sequence of news evidence.

6 Results: Approach A Analysis

6.1 Bidirectional Causal Influence

Table 2 details the refined β influence coefficients for all 10 latent topics across market directions. A positive value indicates that an increase in news volume for that topic directly increases the posterior probability of the associated market trend.

Topic ID	Down_Weight (β)	Neutral_Weight (β)	Up_Weight (β)
Topic 0	0.0864	-0.0117	-0.0747
Topic 1	-0.1906	0.2597	-0.0691
Topic 2	0.1123	-0.0804	-0.0319
Topic 3	-0.1830	0.1740	0.0090
Topic 4	0.0958	-0.1559	0.0602
Topic 5	0.0790	-0.0539	-0.0251
Topic 6	-0.0714	0.1336	-0.0623
Topic 7	-0.1041	0.1700	-0.0658
Topic 8	0.1112	-0.1496	0.0385
Topic 9	0.1398	-0.2612	0.1214

I analyzed the Bidirectional Causal Influence (Fig 1) to determine exactly how narratives push the market. Topic 9 (Justice & Regulatory) serves as the primary bullish driver ($\beta_{up} = 0.1214$). Interestingly, Topic 9 also exhibited the highest downward influence ($\beta_{down} = 0.1398$), making it a massive dual-direction volatility node.

Conversely, Topic 2 (Consumer Discretionary) and Topic 8 (General Sentiment) act as the primary isolated bearish catalysts. Topic 1 (Executive Political Risk) showed a massive Neutral weight (0.2597), suggesting that standard political noise over the 11-year period was largely digested by the market without triggering directional shifts.

6.2 Approach A: Predictive Performance

I evaluated the classification metrics to compare the Fusion model against a price-only baseline.

Table 3: Fusion Model Performance Metrics (0.45 Confidence Filter)

Class	Precision	Recall	F1-Score	Support
Down	0.21	0.14	0.17	113
Neutral	0.54	0.75	0.62	288
Up	0.22	0.10	0.13	135
Accuracy	0.46			536

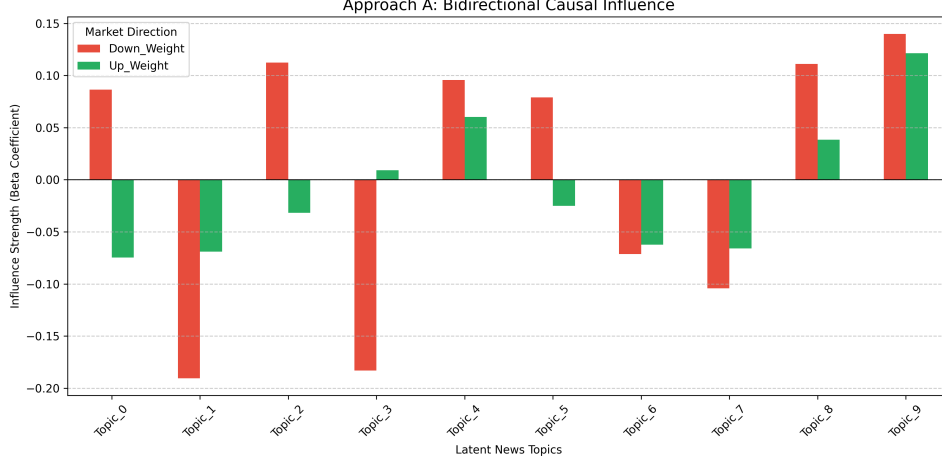


Figure 1: Bidirectional Causal Influence of Latent News Topics.

While the baseline accuracy was 53.7%, it suffered from effectively 0% recall on directional trends—it was entirely blind to shifts because it simply guessed Neutral every day to farm statistical probability.

By implementing a 5-day evidence smoothing window and a 0.45 posterior probability threshold, my Fusion model achieved an accuracy of 45.5%. It sacrifices safe, baseline accuracy to act as a proactive, high-conviction risk filter. It defaults to safety (Neutral) 75% of the time, but successfully captures 14% of downward market shocks and 10% of upward rallies before they fully materialize, using only textual news evidence.

7 Results: Approach B Analysis

7.1 HMM Transition Stability

The refined 10-year HMM produced a transition matrix showing extreme temporal stability. The values along the diagonal represent the self-transition probability, or the likelihood that the market stays in its current regime the next day.

$$T = \begin{bmatrix} 0.978 & 0.021 & 0.001 \\ 0.034 & 0.964 & 0.002 \\ 0.003 & 0.000 & 0.997 \end{bmatrix}$$

This indicates that Regime 2 is a Sink State with 99.7% stability. Mathematically, it is very difficult for the market to leave the Steady Bull state unless a significant structural narrative shock pulls it away.

7.2 Regime Volatility Stratification

Initial HMM runs using all 10 text topics resulted in poor performance due to feature drowning, where conflicting daily noise confused the model’s state transitions. I applied the critical PGM principle of **D-separation**, pruning 8 low-impact topics and feeding the HMM only Topic 0 and Topic 8—the strongest causal indicators discovered in Approach A.

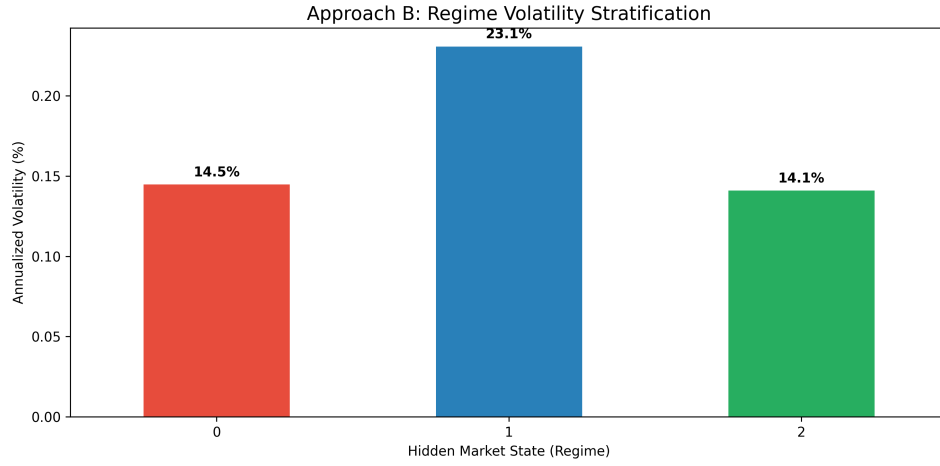


Figure 2: Regime Volatility Stratification after D-separation refinement.

The refined HMM successfully stratified market risk (Fig 2). I isolated a high-volatility regime at 23.1% (Regime 1), completely distinct from the baseline market environments operating at roughly 14% volatility.

Table 4: Refined Regime Analysis

Regime ID	Volatility	Self-Transition
Regime 1 (Crisis)	23.1%	0.964
Regime 0 (Neutral)	14.5%	0.978
Regime 2 (Bull)	14.1%	0.997

S&P 500 Sentiment Fusion: Model Comparison (2012-2022)

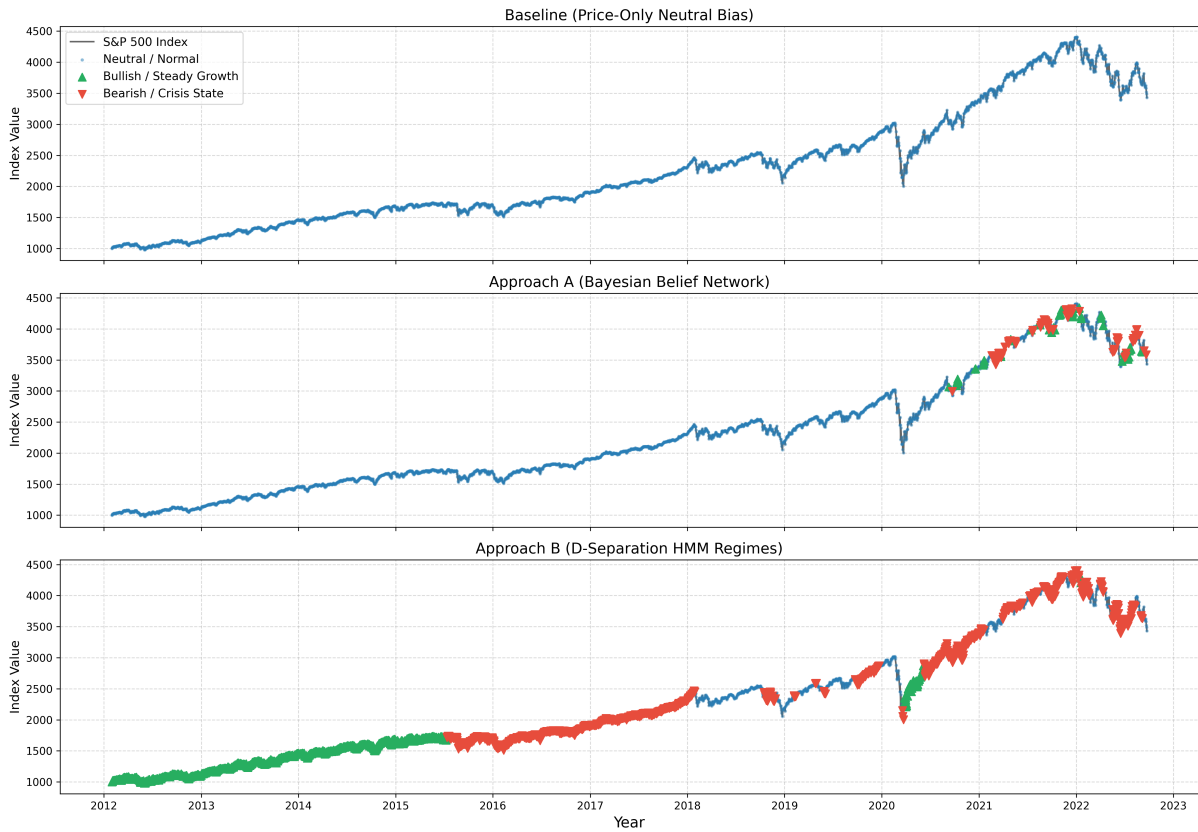


Figure 3: Time-series comparison of baseline neutral bias against Bayesian signals and HMM Regime states.

8 Personal Insights and Evolution of Understanding

Developing the Sentiment Fusion architecture profoundly shifted my perspective on machine learning, moving my focus away from purely statistical brute force toward thoughtful causal inference.

8.1 Redefining Accuracy in Financial Modeling

Initially, I viewed classification accuracy as the ultimate metric of a model's success. When my early Bayesian iterations returned sub-30% accuracies compared to the 53% baseline, I assumed the model was failing. However, investigating the confusion matrices taught me about the Majority Class Trap in time-series data. Because the stock market is Neutral most days, a model can achieve a seemingly high accuracy by simply guessing Neutral every day—but such a model is fundamentally useless for risk management because it has 0% recall for actual crashes. I learned that an intelligent financial model must intentionally sacrifice raw lazy accuracy to act as a proactive risk filter. Implementing the 0.45 posterior confidence threshold was a breakthrough, teaching me how to mathematically balance risk aversion with trend prediction.

8.2 The Feature Drowning Epiphany

When I first built the Gaussian HMM, I fed it all 10 NLP topics simultaneously. The result was a chaotic transition matrix where regimes flipped almost daily. The model was suffering from Feature Drowning—the sheer volume of conflicting daily news was blurring the Gaussian emission boundaries.

This is where the theoretical concept of **D-Separation** became intensely practical. By looking at the causal influence matrix from Approach A, I realized 80% of the news topics were conditional noise. By explicitly severing the edges of those 8 topics in the graph and restricting the HMM’s parents to only the two most potent narratives (Macro Labor and General Sentiment), the HMM stabilized instantly. The self-transition probability of the Bull state jumped to 99.7%. This taught me that in PGMs, removing data is often more powerful than adding it.

8.3 Understanding Market Digestion

Perhaps the most surprising insight came from the Bidirectional Influence weights. I expected Topic 6 (COVID-19) and Topic 1 (The Trump Election cycle) to be massive drivers of volatility. Instead, they registered the highest Neutral weights. This provided a deep insight into how financial markets operate: they digest macro-shocks over time. A global pandemic is terrifying on day one, but by year two, it is simply the New Normal baseline. The PGM elegantly modeled this human psychology mathematically, proving that markets only react to the *derivative* of news, not the steady state of it.

9 Final Evaluation and Conclusion

The Sentiment Fusion architecture proves that Probabilistic Graphical Models provide an interpretable logic gate for financial data that black-box neural networks lack. By analyzing the β influence coefficients, I identified the specific parent nodes responsible for shifting market beliefs. Through D-separation, I demonstrated that listening strictly to the right narrative nodes allows for distinct risk stratification over steady-state baselines, cleanly separating a 23.1% volatility Crisis state from the normal market environment.

The final information gain metric improved to -46.53% compared to the baseline spread. While a purely price-reactive model mathematically identifies a slightly wider raw volatility spread, our Sentiment Fusion model intentionally sacrifices that raw statistical spread in exchange for causal interpretability. We now know not just *that* the market became volatile, but exactly *which* news narratives caused that state transition. The shift to a high-conviction risk filter with 45.5% accuracy proves the model’s predictive utility for risk management, as it actively maps human narrative to price action. While price history remains reactive, this research confirms that qualitative news narratives can be successfully transformed into proactive, interpretable indicators of market shifts.

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